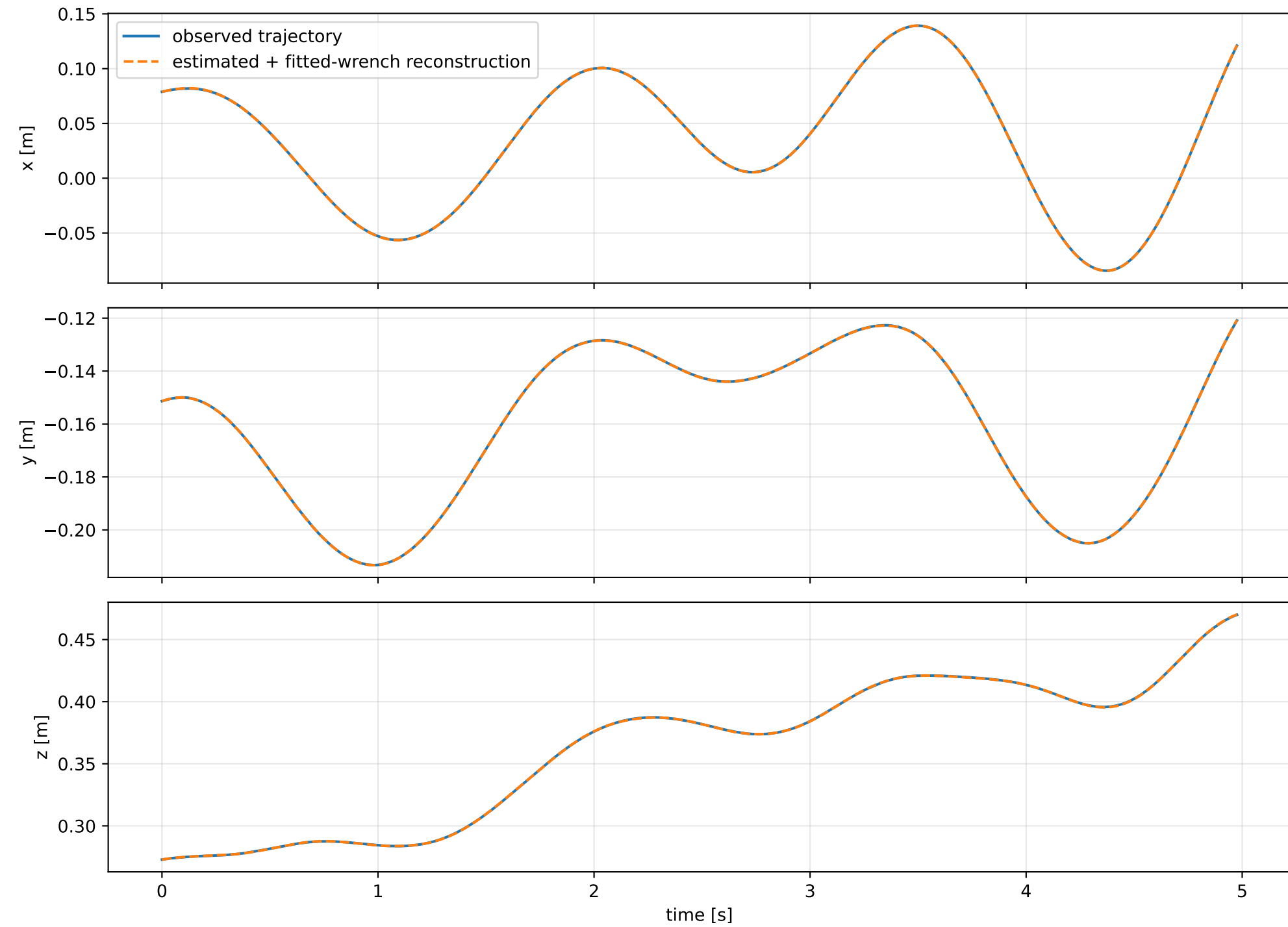
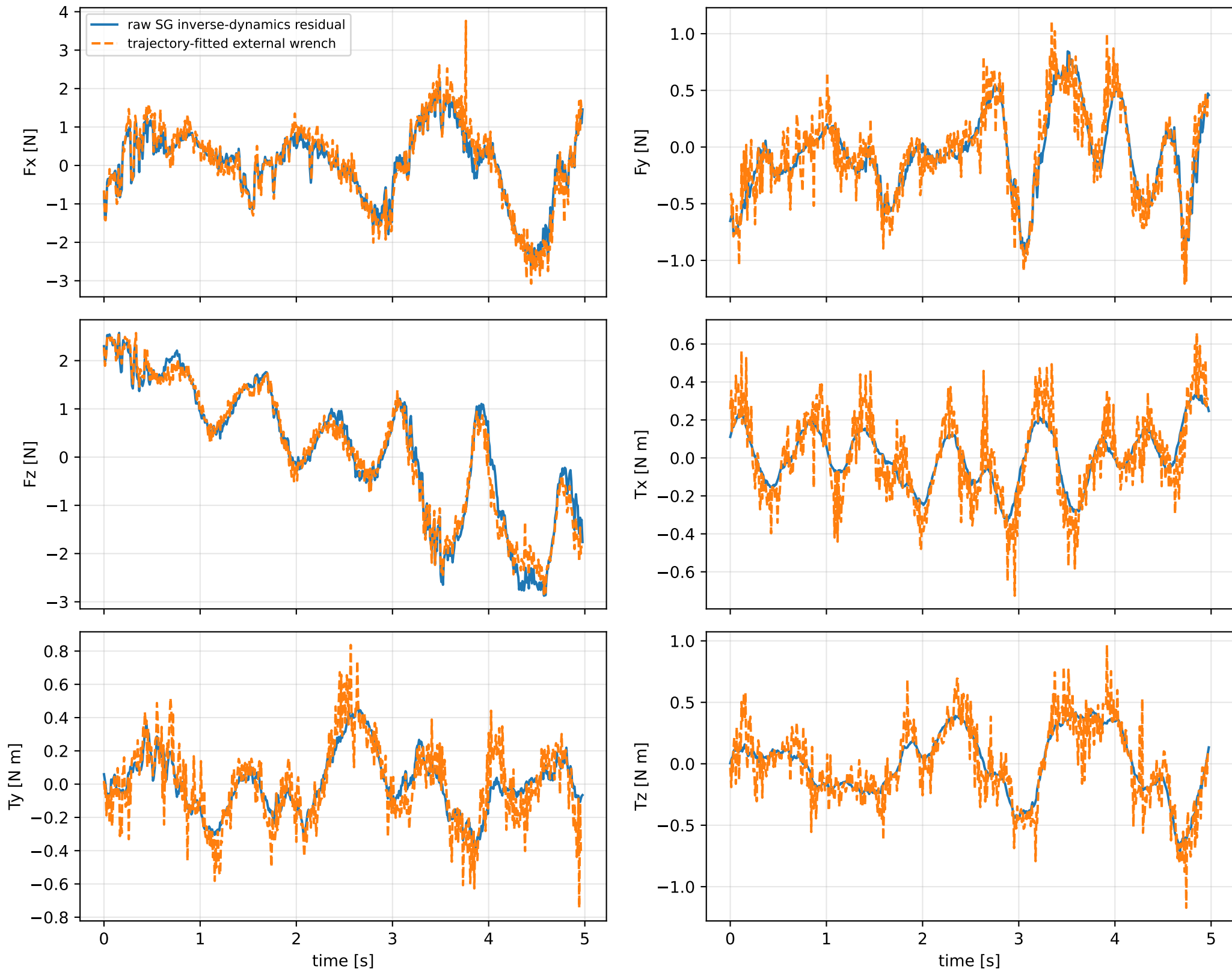


sweep_sg_window_covariance__001: trajectory

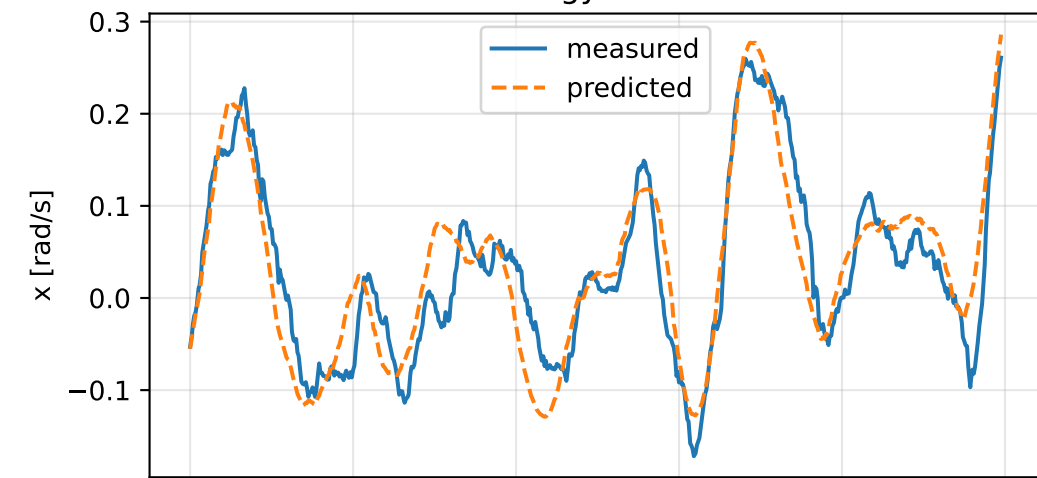


sweep_sg_window_covariance_001: residual wrench

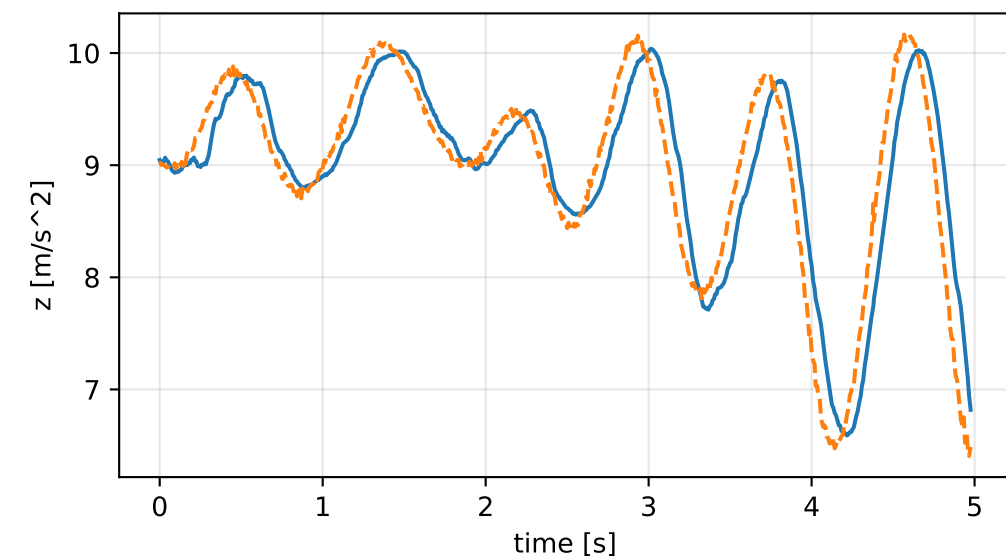
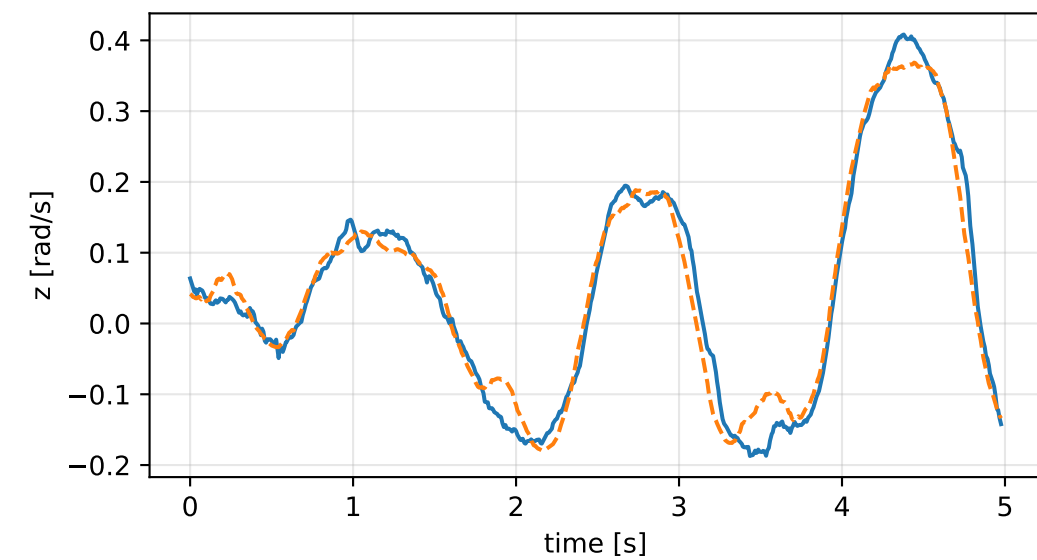
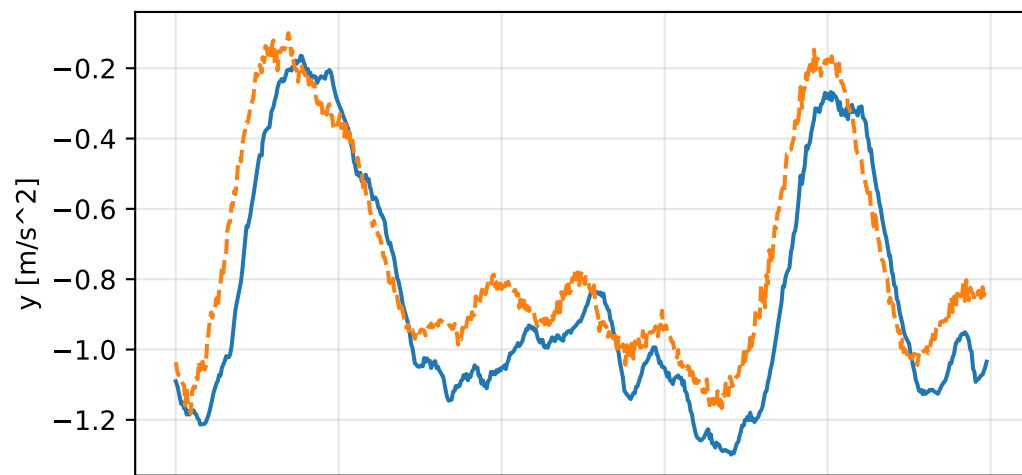
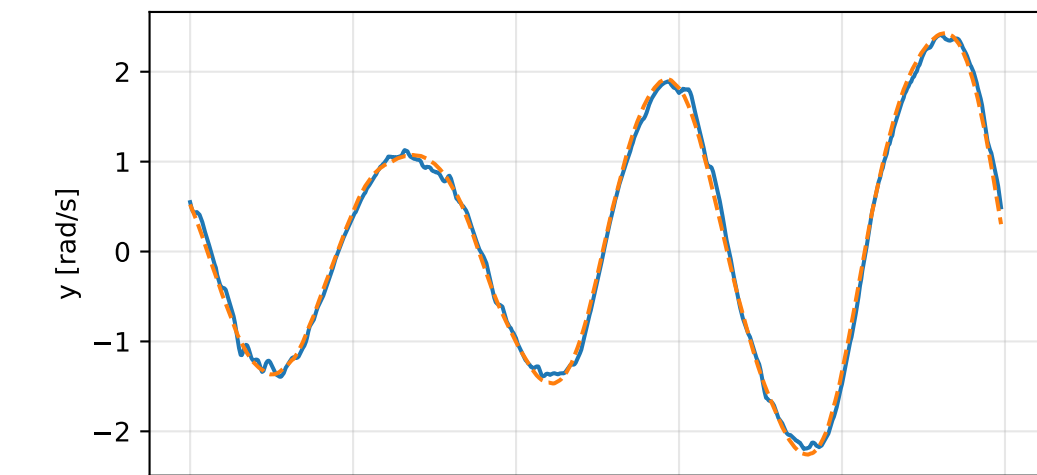
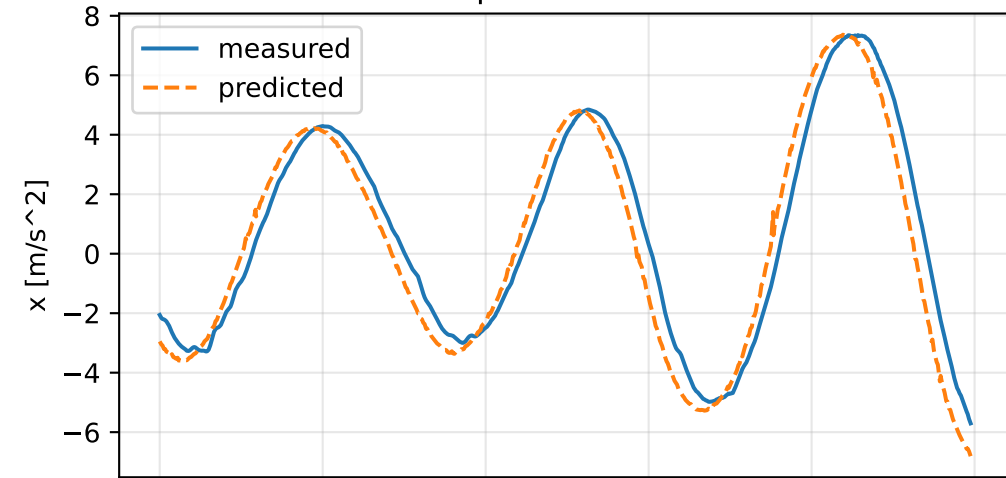


sweep_sg_window_covariance_001: measured sensor comparison

gyro



specific force



sweep_sg_window_covariance__001: nominal -> estimated parameters

Case: sweep_sg_window_covariance__001
status: completed (all stages completed)

mass [kg] nominal 2.35159759 estimated 3.88179174 delta 1.53019415

inertia [kg m^2] (nominal -> estimated; delta)

[6.5000061e-02 -7.2789925e-07 1.9015080e-05] -> [0.3504662 0.0006446 -0.0751233] d=[0.2854662 0.0006453 -0.0751424]
[-7.2789925e-07 6.4952656e-02 5.9167305e-08] -> [6.4458253e-04 3.0015301e-01 -6.4605775e-05] d=[6.4531043e-04 2.352
[1.9015080e-05 5.9167305e-08 1.2899211e-01] -> [-7.5123335e-02 -6.4605775e-05 6.1030042e-01] d=[-7.5142350e-02 -6.46649

CoG [m] [-2.0247086e-03 -3.0526579e-05 9.5097496e-03] -> [0.0111581 -0.0026782 -0.0435313] d=[0.0131828 -0.0026477

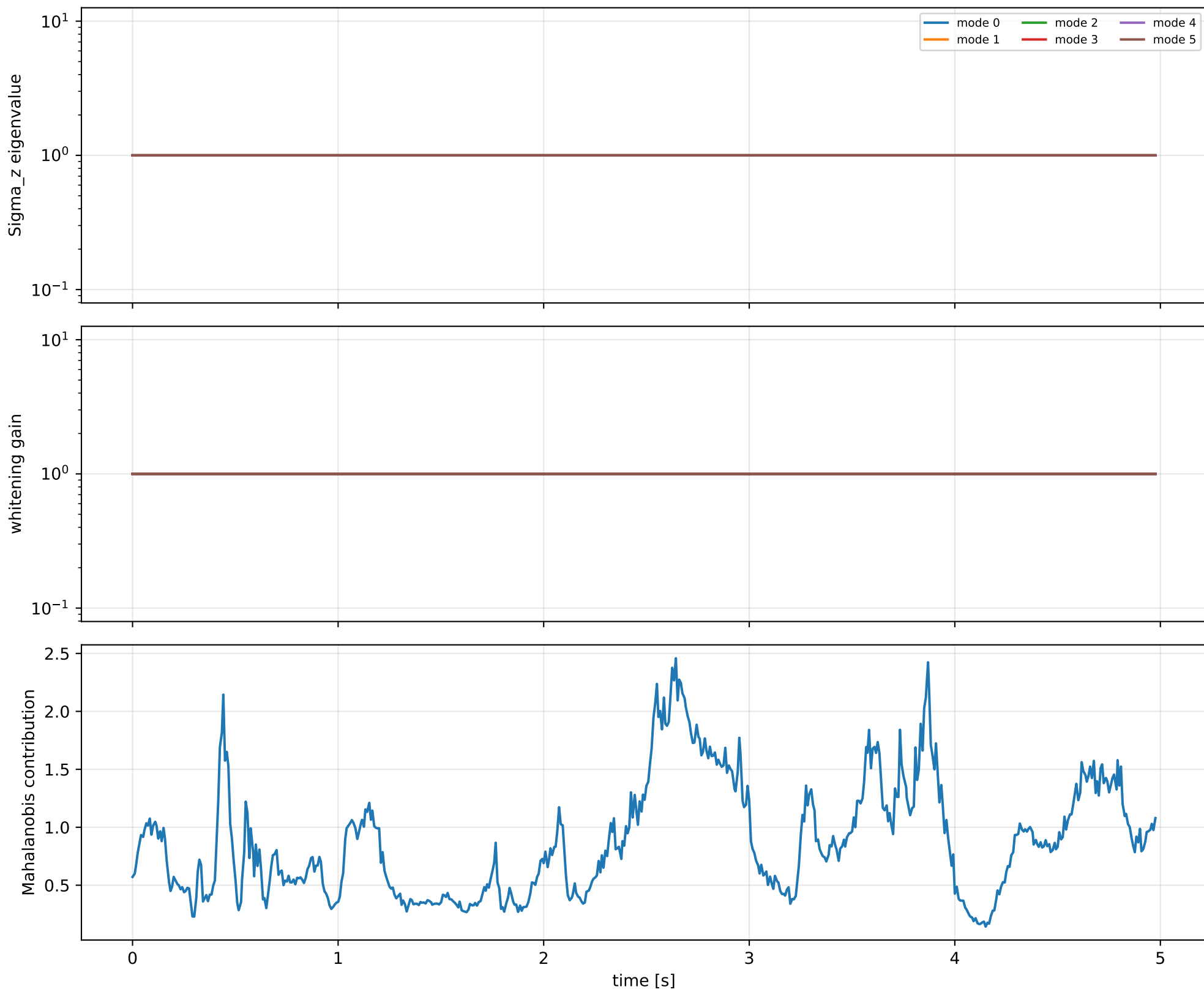
rotor force effectiveness

[1. 1. 1. 1.] -> [1.3325313 1.6200218 1.429952 1.2570383] d=[0.3325313 0.6200218 0.429952 0.2570383]

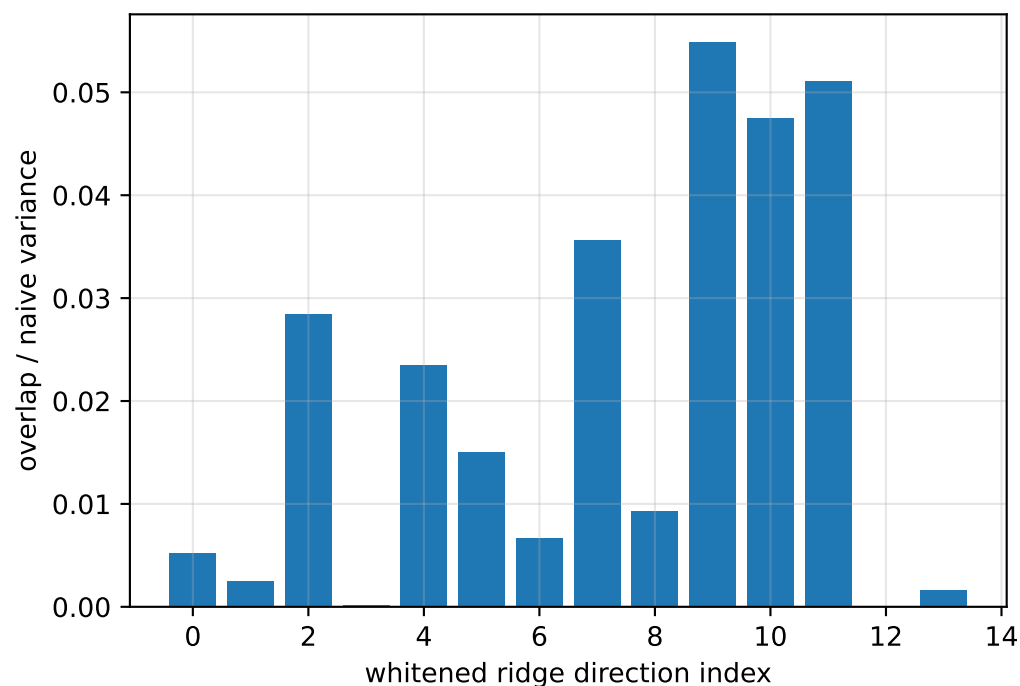
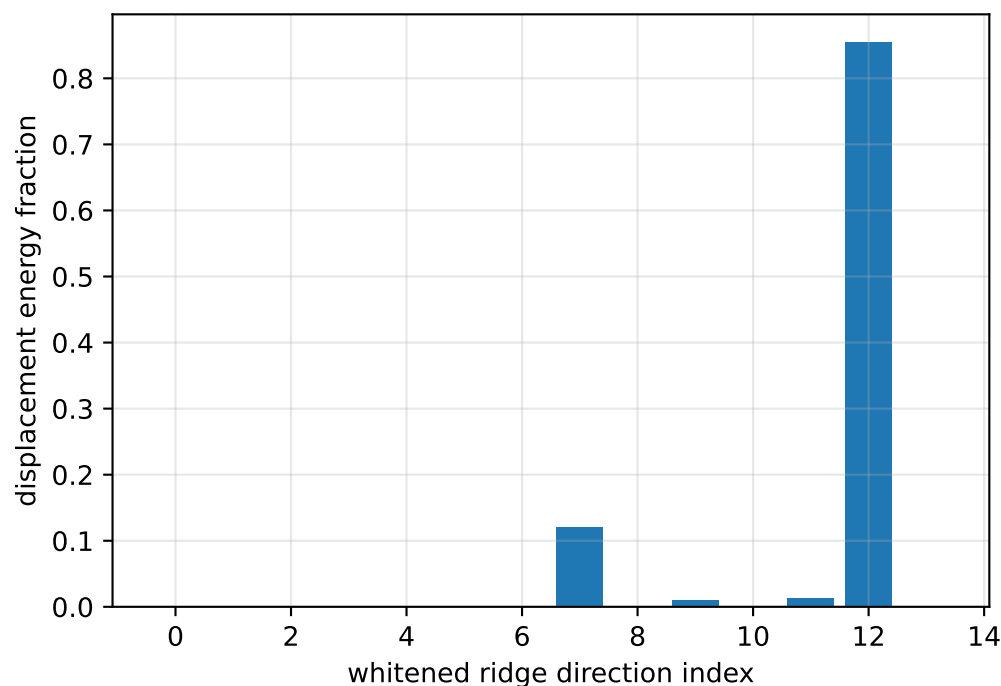
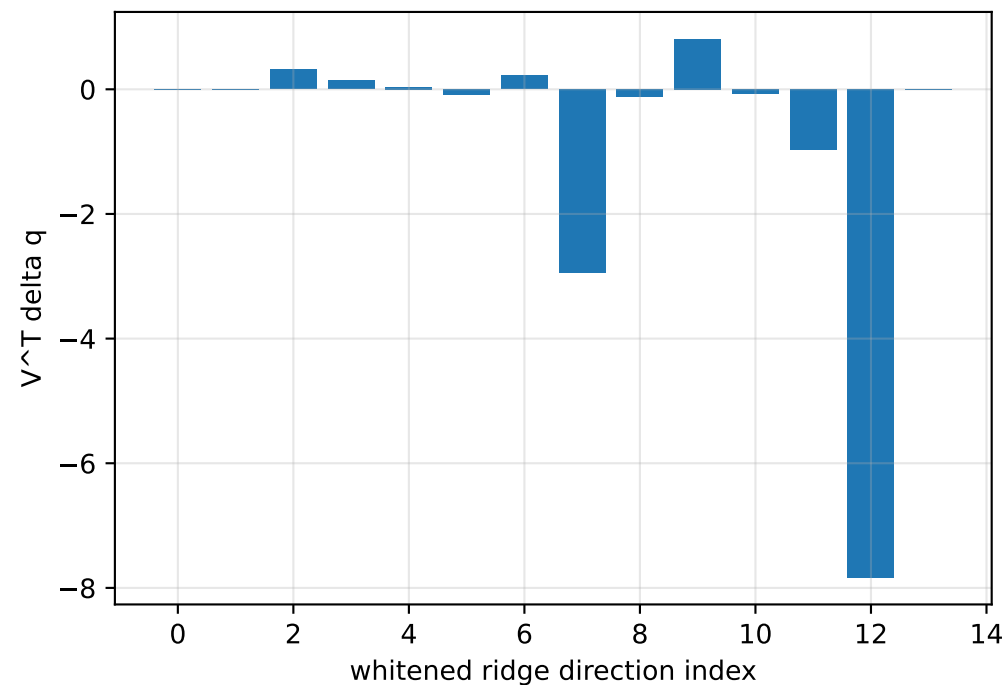
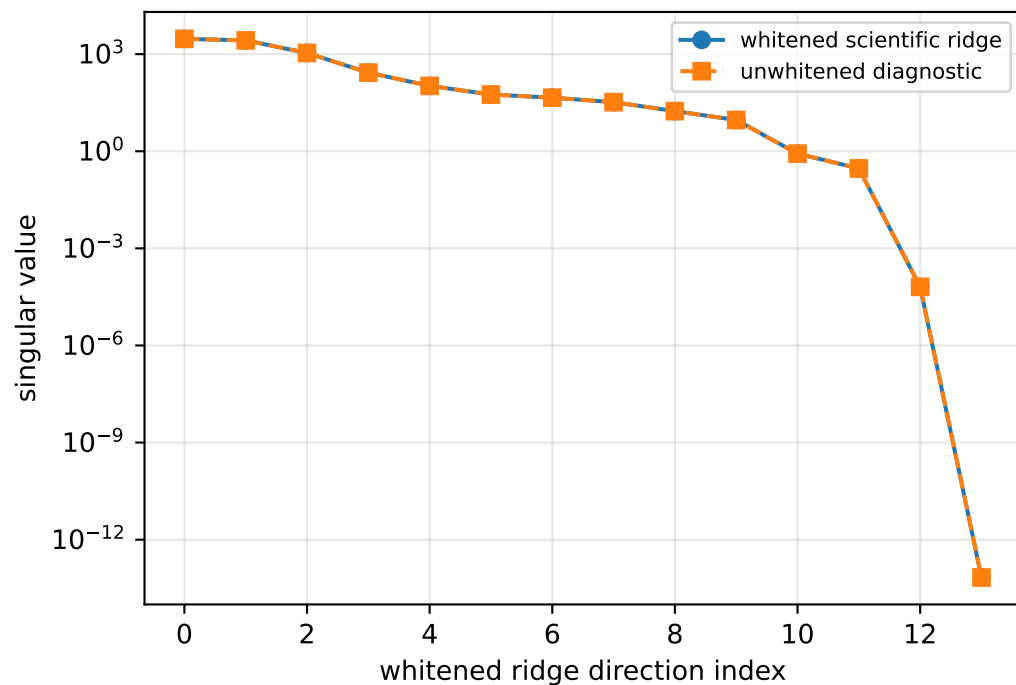
rotor lag [s] 0.19500227

gimbal lag [s] 0.0438094006

sweep_sg_window_covariance__001: covariance weighting

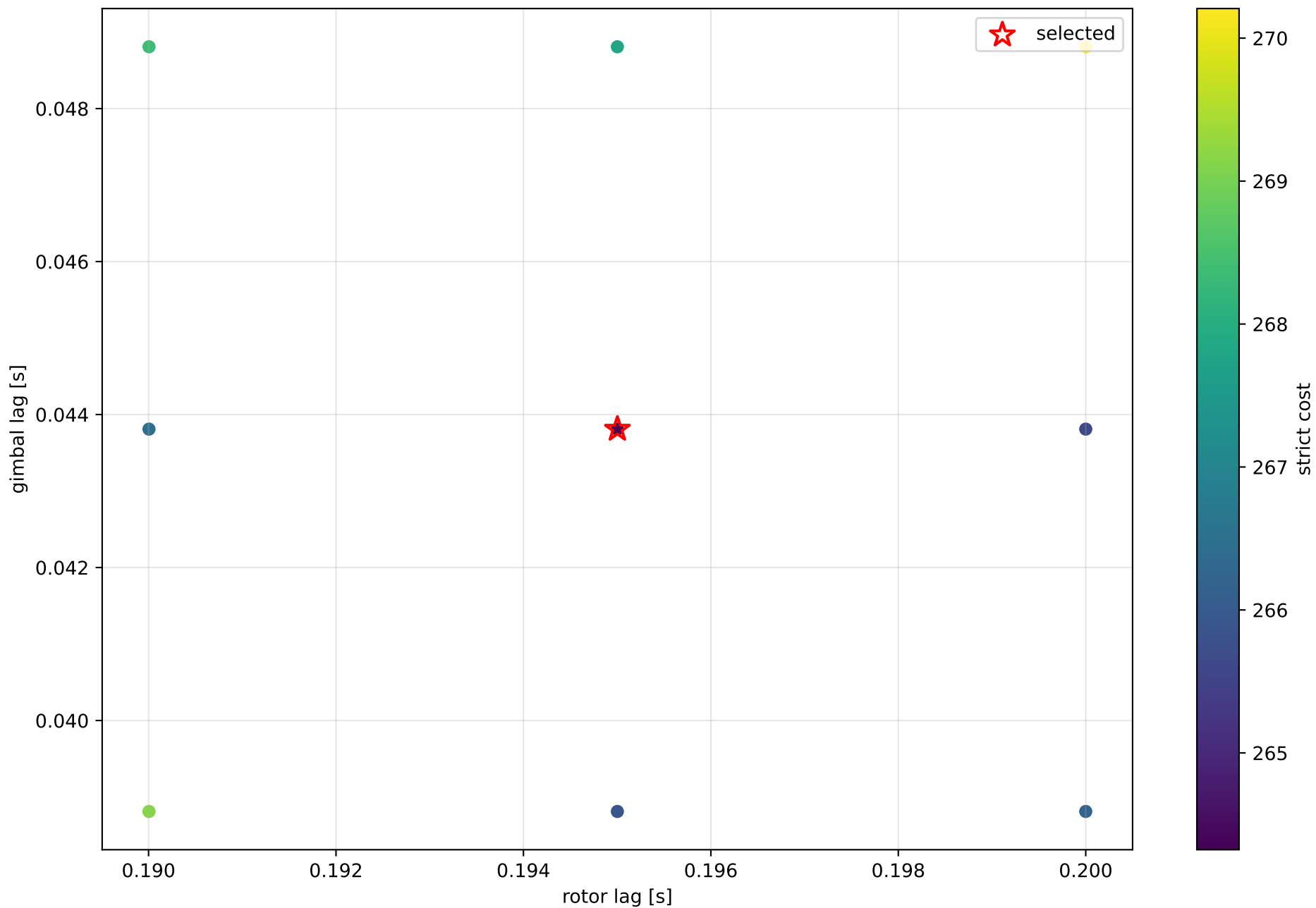


sweep_sg_window_covariance_001: parameter ridge



exact scale gauge: $||v_scale||=1.9759e-13$, rank=13, nullity=1

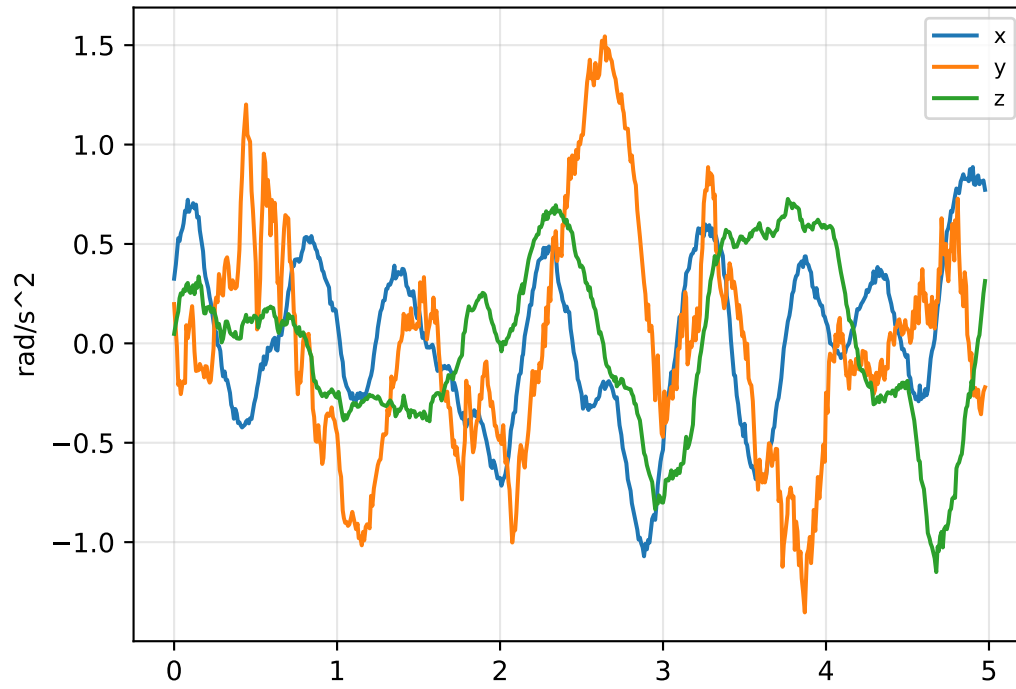
sweep_sg_window_covariance__001: strict lag candidates



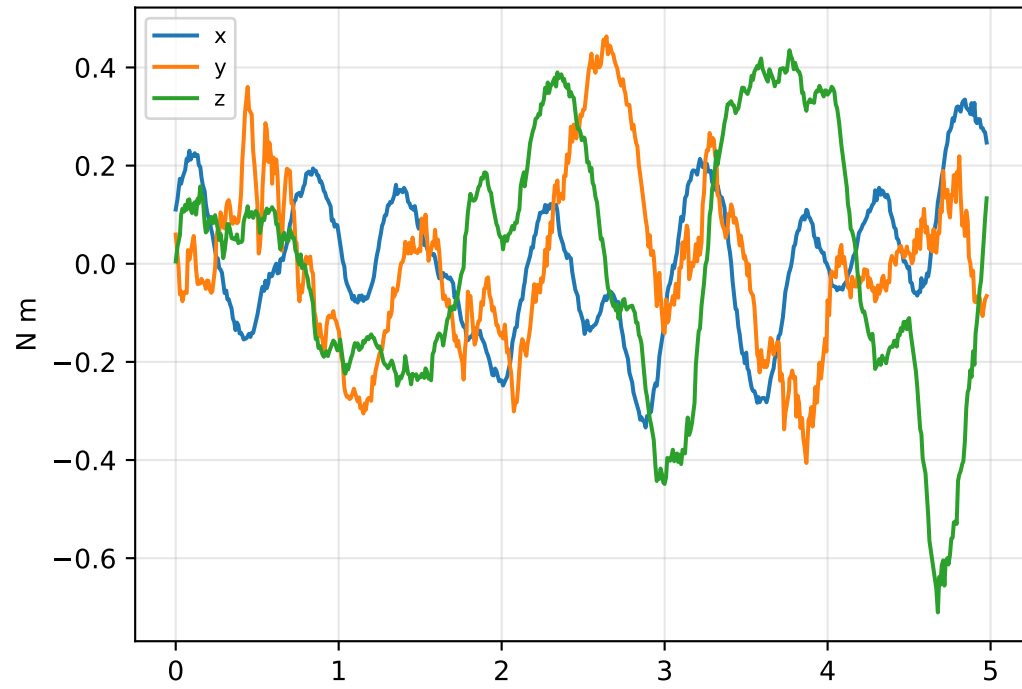
boundary flags: rotor lower=False upper=False; gimbal lower=False upper=False

sweep_sg_window_covariance__001: acceleration/wrench closure

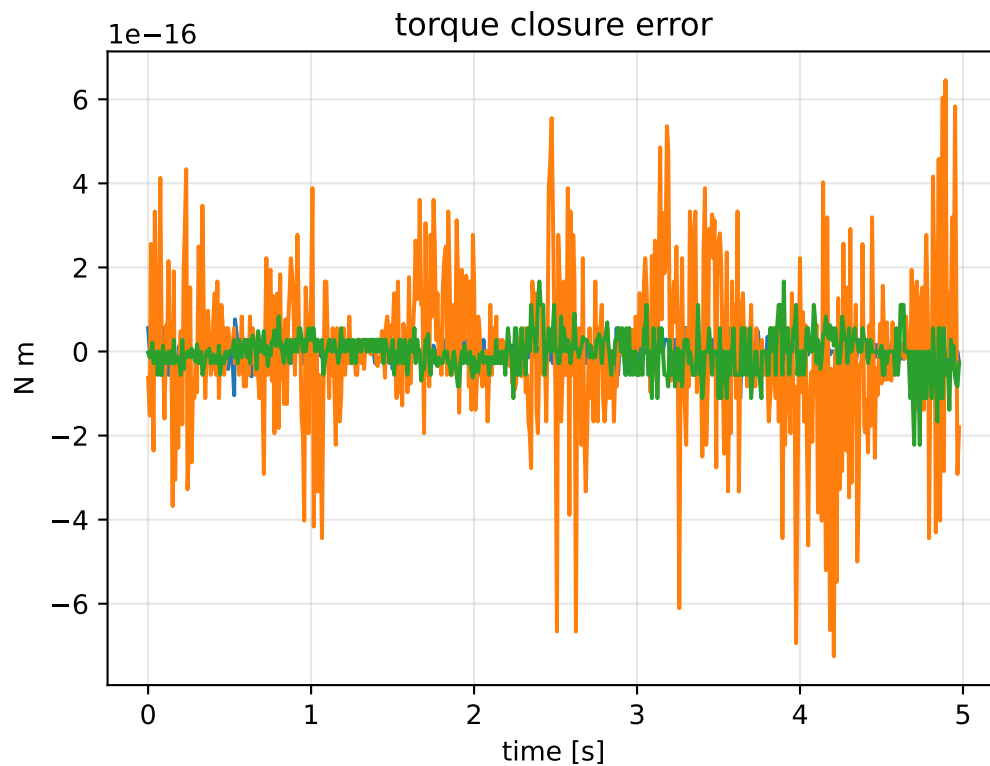
angular-acceleration residual



raw torque residual



torque closure error



principal moments [kg m²]
 nominal: [0.064953 0.065 0.128992]
 estimated: [0.300141 0.330322 0.630457]
 estimated / nominal: [4.620916 5.081876 4.88756]

force closure max/rms: 1.11022e-14 / 2.40478e-15
 torque closure max/rms: 7.25114e-16 / 1.11498e-16